
EE 708 : Information Theory and Coding
IIT Bombay, Dept. of EE, Semester: Spring 2023-2024
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Homework 1

Exercises on information measures, their properties, and tools.

From the book by Cover and Thomas, Chapter 2 exercises:

2.2, 2.5, 2.6, 2.12, 2.14, 2.19, 2.27, 2.28, 2.39

Additional questions

Question: 1 (i) Solve Ash, ex. 1.5: Use $D(p||q) \geq 0$ to prove the AM-GM inequality.

(ii) Use Jensen's inequality to prove the AM-GM inequality for positive numbers.

$$\prod_{i=1}^n x_i^{a_i} \leq \sum_{i=1}^n a_i x_i.$$

where $a_i \geq 0$ for all i , and $\sum_i a_i = 1$. What is the condition for equality?

Question: 2 Prove the data processing inequality for KL divergence: If $W_{Y|X}$ is a stochastic matrix (conditional probability matrix), then for any two distributions p_X, q_X , $D(p_Y||q_Y) \leq D(p_X||q_X)$. Here p_Y (resp. q_Y) denotes the marginal distribution of Y under p_X (resp. q_X) and $W_{Y|X}$.

Question: 3 Let $p = (p_1, p_2, \dots, p_M)$ be a probability mass function, and W be a doubly stochastic matrix, i.e. a square matrix of non-negative elements with row and column sums equal to 1. Show that

$$H(pW) \geq H(p)$$

for every distribution p on $\{1, 2, \dots, M\}$. **Hint.** Use Data processing inequality for KL divergence.

Question: 4 (Teaser) Let $p = (p_1, p_2, \dots, p_M)$ and $q = (q_1, q_2, \dots, q_M)$ be two distributions, with ordering $p_1 \geq p_2 \geq \dots \geq p_M$, and $q_1 \geq q_2 \geq \dots \geq q_M$. q is said to be 'more uniform' than p , and written as $q > p$ if

$$\sum_{i=1}^k q_i \leq \sum_{i=1}^k p_i$$

for $k = 1, 2, \dots, M$. Show that if $q > p$, then there exists a doubly stochastic matrix W , i.e. a square matrix of non-negative elements with row and column sums equal to 1, such that $q = pW$.